

Session – 9 Exercise SVM

Topic: Support Vector Machines (SVM)

Dataset: Wine Dataset — `load_wine` (from `sklearn.datasets`)

Input: 178 samples, 13 numeric predictors

Target Variable: wine cultivar class (0, 1, 2 - three classes)

Q1. What is a Support Vector Machine (SVM), and what problem is it solving? Explain the concepts of the *maximum-margin hyperplane* and *support vectors*. Why does maximizing the margin between classes tend to produce a classifier that generalizes well to unseen data, compared to any hyperplane that merely separates the training points?

Q2. Load and explore the Wine dataset. Report the shape of the data, the list of feature names, and the class balance of the target (counts and percentages). Given that SVM was originally formulated as a binary classifier, briefly explain how it is extended to handle more than two classes (one-vs-rest or one-vs-one).

Q3. Split the data into training (80%) and test (20%) sets using `train_test_split` with `random_state=42` and `stratify=y`. Scale the features using `StandardScaler`, fitting it on the training data only. Why is feature scaling especially important for SVMs (more so than for a model like decision trees)? What could go wrong if one feature were left on a much larger numeric scale than the others?

Q4. Fit a linear-kernel SVM (`SVC(kernel='linear')`) on the scaled training data. Report how many support vectors were selected (`model.n_support_`) and what fraction of the training set they represent. What does it mean, geometrically, that only a subset of the training points becomes support vectors while the rest have no influence on the decision boundary?

Q5. Generate predictions on the test set, build the confusion matrix, and compute accuracy, precision, recall, and F1-score (using macro-averaging, since this is a 3-class problem). Which class, if any, is hardest for the model to distinguish? Support your answer with numbers from the confusion matrix.

Q6. The kernel trick and the C hyperparameter. Explain what the 'kernel trick' allows an SVM to do that a plain linear decision boundary cannot, and describe in plain terms what the linear, polynomial, and RBF (Gaussian) kernels each do. Separately, explain the role of the regularization parameter C: what happens to the margin and to training error as C is increased vs. decreased?

Q7. Compare kernels with GridSearchCV. Using 5-fold cross-validation on the training set, tune C for a linear kernel and tune both C and gamma for an RBF kernel (`SVC(kernel='rbf')`). Report the best parameters and best cross-validation accuracy for each kernel, then evaluate both final models on the held-out test set in a comparison table. Which kernel would you recommend for this dataset, and why?

Q8. Visualize the decision boundary. Using only two features (e.g. 'alcohol' and 'color_intensity') and `make_moons` or the two chosen Wine features, plot the decision regions produced by a linear-kernel SVM side by side with an RBF-kernel SVM. What does the shape of each boundary reveal about the flexibility of the two kernels, and what is the risk of choosing a kernel that is too flexible for the amount of training data available?